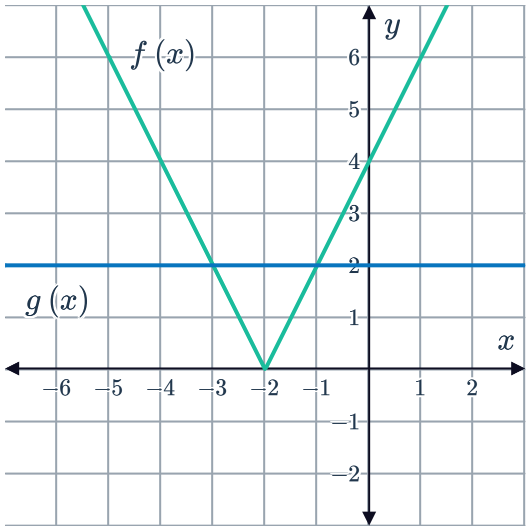
Solve equations graphically

1 3

2

a



b 2

c $x=-1,-3$

3

a 3 solutions

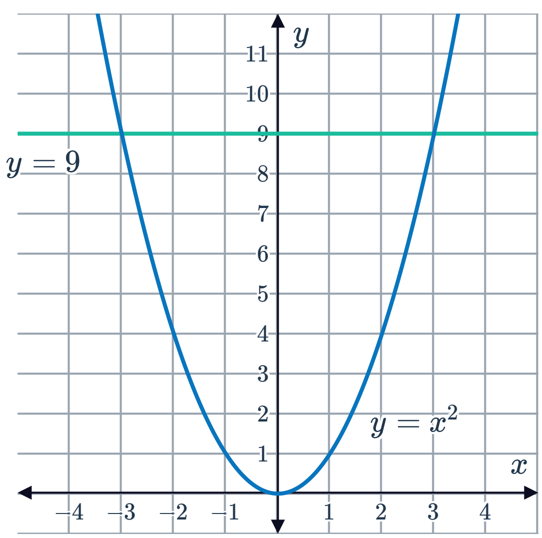
b $x=3,2,-1$

c 1 solution

d 2 solutions

4

a

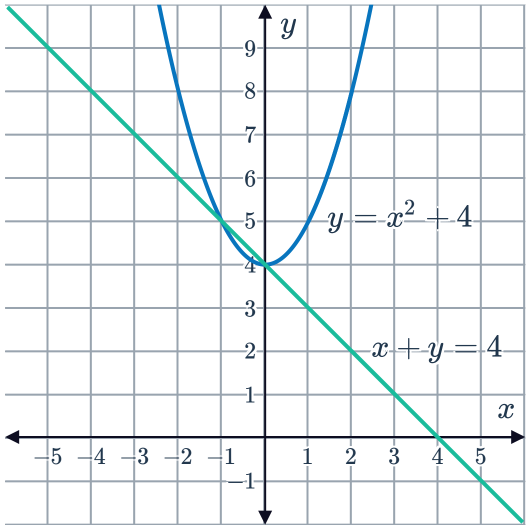


b $(-3,9)\,,\,(3,9)$

c $x=3,-3$

5

a

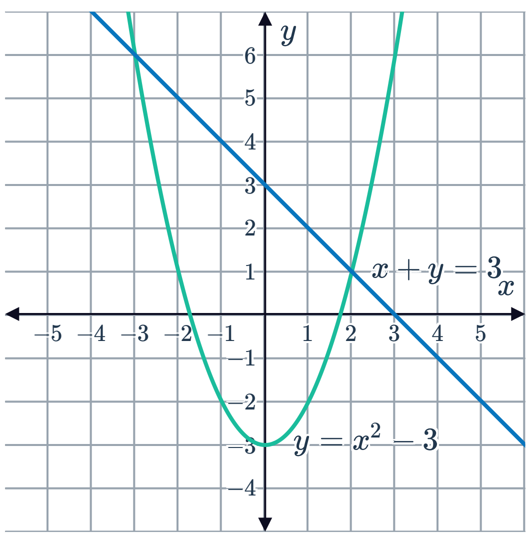


b 2 solutions

c $(0, 4)$, $(-1, 5)$

6

a

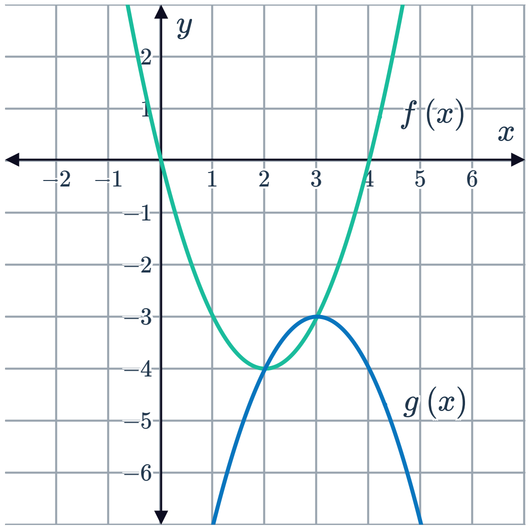


b 2 solutions

c $(2, 1)$, $(-3, 6)$

7

a



b $(3, -3), (2, -4)$

c $x = 2, 3$

Solve equations using technology

8

a $x = 8.37$

b $x = -5$

c $x = 1, -1.28$

d $x = -0.44$

e $x = 0.71$

f $x = 0.55$

g $x = -4.71$

h $x = 3$

i $x = 3, -9$

j $x = 0, 6.73, -0.40$

k $x = 1.06$

l $x = 4, -3$

m $x = -7, 9, 0, -9$

n $x = 0.36$

9 $x = -0.562$

Applications

10

a $h\left(x\right)=x^3+3x-7+2^x$

b

t	-2	-1	0	1	2
$h\left(x\right)$	$-\frac{83}{4}$	$-\frac{21}{2}$	-6	-1	11

c $1<x<2$

11

a $215t^2-642t-10$

b

t	3	3.25	3.5	3.75	4
$h\left(t\right)$	-1	174.44	376.75	605.94	862

c A value of t that is slightly greater than 3.

d The two quantities being measured here are time and population, in which case neither can be negative.